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Time to Practice: Applying Unsupervised Learning to Supervised Learning Tasks
Graded Assignment • 15 min

Due Oct 28, 11:59 PM IST

Your grade: 100%

Your latest: **100%** • Your highest: **100%** • To pass you need at least 80%. We keep your highest score.

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1. A real estate pricing model is becoming less accurate as property prices in a new district are significantly higher due to recent developments. What phenomenon does this scenario illustrate? 1 / 1 point

- Model overfitting
- Covariate shift
- Algorithm bias
- Data leakage

Nice work

Covariate shift refers to the situation where the distribution of input variables changes, while the output variable distribution remains the same. In this case, the shift in property prices in a new district represents a covariate shift, affecting the model's accuracy.

2. In a scenario where a dataset contains hidden subgroups within the classes, which unsupervised learning method could be used to discover these subgroups to improve the performance of a supervised classifier? 1 / 1 point

- Feature selection based on information gain
- Clustering
- Outlier detection

Nice work

Clustering can be used to identify hidden subgroups or clusters within the data. Understanding these subgroups can help in building more accurate and tailored classifiers in supervised learning, possibly for each identified subgroup.

3. In a scenario where a biologist wants to classify long RNA sequences, what approach could be used to make the classification task more manageable? 1 / 1 point

- Enhancing sequence resolution
- Developing an embedding
- Increasing sequence length

Nice work

Developing an embedding that maps long RNA sequences to a lower-dimensional space can simplify the classification task, making it feasible to apply more sophisticated classification techniques that might not work in a high-dimensional space.

4. What is the benefit of using multiple classifiers in a scenario where data points are clustered into subgroups? 1 / 1 point

- Reduced computational cost
- Enhanced classification accuracy
- Decreased training time

Nice work

Using multiple classifiers allows for specialized handling of different subgroups identified by clustering, leading to more accurate classification of data points based on their characteristics.

5. What is a key challenge in supervised learning that manifold learning can help address? 1 / 1 point

- Overfitting models
- Finding non-linear structure, especially in high-dimensional data
- Insufficient training examples

Nice work

Manifold learning can help in discovering the non-linear structure in the data, especially in high dimensions, which can be crucial for improving the performance of supervised learning models.

6. What are valid reasons for data drift/covariate shift to happen? (Select all that apply.) 1 / 1 point

- [x] A previously reliable feature is lost or becomes unreliable

Nice work

Any significant change in the distribution of the input variables between training and test can be considered data drift/covariate shift.

- [x] The training data was sampled from an unrepresentative group

Nice work

Any significant change in the distribution of the input variables between training and test can be considered data drift/covariate shift.

- [ ] Access to the data suddenly becomes more slower and/or more costly
- [x] Monthly or seasonal effects

Nice work

Any significant change in the distribution of the input variables between training and test can be considered data drift/covariate shift.